



DISCOVERY OF A NOVEL Gene USING ALDH2

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Lab Section 04/03/2026

INTRODUCTION

- Aldehyde Dehydrogenase 2 (ALDH2)
- Homo Sapiens
- Accession Number: NP_000681.2
- Chosen due to our interest if the same or similar gene existed in other species

TBLASTN

- TBLASTN was used from the National Center for Biotechnology Information (NCBI) website
- No default parameters were modified
- Specifically searched the “Whole-genome contigs shotgun” (wgs) database
- Taxa: *Microcebus Ravelobensis*, otherwise known as the Golden Brown Mouse Lemur, was our organism filter



TBLASTN RESULTS

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
✓	Microcebus ravelobensis isolate RMR55 scaffold36670	761	761	97%	0.0	71.77%	18842	VSMG01154874.1
✓	Microcebus ravelobensis isolate RMR55 scaffold100250	184	184	91%	8e-48	28.90%	6162	VSMG01325304.1
✓	Microcebus ravelobensis isolate RMR55 scaffold43866	134	909	82%	3e-31	92.31%	16318	VSMG01017660.1
✓	Microcebus ravelobensis isolate RMR55 scaffold46447	129	129	28%	2e-29	43.33%	15528	VSMG01300566.1
✓	Microcebus ravelobensis isolate RMR55 scaffold123864	119	119	43%	4e-26	34.06%	4077	VSMG01057806.1
✓	Microcebus ravelobensis isolate RMR55 scaffold19386	119	332	41%	5e-26	50.40%	28872	VSMG01291763.1
✓	Microcebus ravelobensis isolate RMR55 scaffold6340	103	352	44%	6e-21	69.70%	49013	VSMG01001100.1
✓	Microcebus ravelobensis isolate RMR55 scaffold18105	90.1	289	38%	8e-17	68.25%	30032	VSMG01283278.1
✓	Microcebus ravelobensis isolate RMR55 scaffold6641	86.3	86.3	36%	1e-15	32.11%	48172	VSMG01315293.1

<https://blast.ncbi.nlm.nih.gov/>

PROTEIN SEQUENCE

[Download](#) [GenBank](#) [Graphics](#) Sort by:

Microcebus ravelobensis isolate RMR55 scaffold43866, whole genome shotgun sequence
Sequence ID: [VSMG01017660.1](#) Length: 16318 Number of Matches: 10

Range 1: 15037 to 15138 [GenBank](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame
70.9 bits(172)	8e-11	Compositional matrix adjust.	33/34(97%)	33/34(97%)	0/34(0%)	-2
Query 87	FQLGSPWRRMDASHRGLLNRLADLIERDRTYLA			120		
	FQLGSPWRRMDAS RGLLNRLADLIERDRTYLA					
Sbjct 15138	FQLGSPWRRMDASSRGLLNRLADLIERDRTYLA			15037		

Range 2: 13820 to 13966 [GenBank](#) [Graphics](#) [Next Match](#) [Previous Match](#) [First Match](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame
65.1 bits(157)	6e-09	Compositional matrix adjust.	31/49(63%)	37/49(75%)	4/49(8%)	-1
Query 121	ALETLDNGKPYVISYLVLDMLVKCLRY-----AGWADKYHGKTIPIDG			165		
	ALETLDNGKPYVISYLVLDMLVKCLRY+ W+ G+ + ++G					
Sbjct 13966	ALETLDNGKPYVISYLVLDMLVKCLRYWFKFVLWSSSAKGEGMLLEG			13820		

Range 10: 2226 to 2390 [GenBank](#) [Graphics](#) [Next Match](#) [Previous Match](#) [First Match](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame
85.5 bits(210)	2e-15	Compositional matrix adjust.	44/61(72%)	46/61(75%)	6/61(9%)	-3
Query 450	TKDLDKANYLSQALQAGTVWVNCYDVFGAQSPFGGYKMSGSGRELGEYGLQAYTEVKTVT			509		
	T P D N + S VNCYDVFGAQSPFGGYKMSGSGRELGEYGLQAYTEVKT+V					
Sbjct 2390	TPSDFCNLISN-----YRVNCYDVFGAQSPFGGYKMSGSGRELGEYGLQAYTEVKTVS			2229		
Query 510	V 510					
Sbjct 2228	T 2226					

https://blast.ncbi.nlm.nih.gov/Blast.cgi?CMD=Get&VIEW_RESULTS=FromRes&RID=WVDSHE5J014&UNIQ_OBJ_NAME=A_SearchRes&uts_1w85qb_2h6N_dwaCrSsqokk_GTMVb_1aG9fk&QUERY_INDEX=0#alnHdr_VSMG01017660

Final Amino Acid Sequence:

87

FQLGSPWRRMDASSRGLLNRLADLIERDRTYLA ALETL
DNGKPYVISYLVLDMLVKCLRY YAGWADKYHGKTIPI D
GDFS SYTRHEPVGVCGQII PWNFP LLMQAWKLGPALATG
NVVVMKVAEQTPLTALYVANLIKEAGFP PGVVNIVPGFG
PTAGAAIASHEDVDKVAFTGSTEIGRLIQVAAGSSNLKR
VTLELGGKSPNIIMSDADV DWAVEQAHFALFFNQGCC
AGSRTFVQEDVYNEFVERS VARAKSRVVG NPFDSKTEQG
PQVDETQFKKILGYINSGKQEGAKLLCGGGPAADRGYYI
QPTVFGDVQDGMTIAKEE IFGPVMQILKFKTIEEVVGRA
NNSNYGLAAAVFTKDL DKANYLSQALQAGTVWVNCYDVF
GAQSPFGGYKMSGSGRELGEYGLQAYTEVKTVST 510



BLASTX

- BLASTX was also used from the NCBI
- None of the default parameters were changed
- The “Non-redundant protein sequences (nr)” database was used validate our protein as truly novel

AMINO ACID TO DNA

Master DNA Sequence Excluding Gaps:

```
ATGTTTCAAT TAGGTTCTCC TTGGCGTCGT ATGGATGCTT CTTCTCGTGG TCGTTTATTA AATCGTTTAG CTGATTTAAT TGAACGTGAT
CGTACTTATT TAGCTGCTTT AGAAACTTTA GATAATGGTA AACCTTATGT TATTTCTTAT TTAGTTGATT TAGATATGGT TTTAAATGT
TTACGTTATT ATGCTGGTTG GGCTGATAAA TATCATGGTA AAACCTATCC TATTGATGGT GATTTTTTTT CTTATACTCG TCATGAACCT
GTTGGTGTTT GTGGTCAAAAT TATTCCTTGG AATTTTCCTT TATTAATGCA AGCTTGGAAA TTAGGTCCTG CTTTAGCTAC TGGAATGTT
GTTGTTATGA AAGTTGCTGA ACAAACCTCT TTAAGTCTT TATATGTTGC TAATTTAATT AAAGAAGCTG GTTTTCCTCC TGGTGTTGTT
AATATTGTTT CTGGTTTGG TCCTACTGCT GGTGCTGCTA TTGCTTCTCA TGAAGATGTT GATAAAGTTG CTTTACTGG TTCTACTGAA
ATTGGTCGTT TAATTCAAGT TGCTGCTGGT TCTTCTAATT TAAAACGTGT TACTTTAGAA TTAGGTGGTA AATCTCCTAA TATTATTATG
TCTGATGCTG ATGTTGATTG GGCTGTTGAA CAAGCTCATT TTGCTTATT TTTAATCAA GGTCAATGTT GTTGTGCTGG TTCTCGTACT
TTTGTTCAAG AAGATGTTTA TAATGAATTT GTTGAACGTT CTGTTGCTCG TGCTAAATCT CGTGTTGTTG GTAATCCTTT TGATTCTAAA
ACTGAACAAG GTCCTCAAGT TGATGAACT CAATTTAAAA AAATTTTAGG TTATATTAAT TCTGGTAAAC AAGAAGGTGC TAAATTATTA
TGTGGTGGTG GTCCTGCTGC TGATCGTGGT TATTATATTC AACCTACTGT TTTTGGTGAT GTTCAAGATG GTATGACTAT TGCTAAAGAA
GAAATTTTGG GTCCTGTTAT GCAAATTTTA AAATTTAAAA CTATTGAAGA AGTTGTTGGT CGTGCTAATA ATTCTAATTA TGGTTTAGCT
GCTGCTGTTT TACTAAAGA TTAGATAAAA GCTAATTATT TATCTCAAGC TTTACAAGCT GGTACTGTTT GGGTTAATTG TTATGATGTT
TTTGGTGCTC AATCTCCTTT TGGTGGTTAT AAAATGTCTG GTTCTGGTCG TGAATTAGGT GAATATGGTT TACAAGCTTA TACTGAAGTT
AAACTGTTT CACTTAA
```

<https://www.cusabio.com/Protein-to-DNA-Sequence-Generator.html?srsId=AfmBOorYVCExPVCUd1DZRPjCtHLvPhHBbtQNNUQs8ftv9TytxT0w4Llm>

Sequences producing significant alignments

Download ▾

Select columns ▾

Show

50 ▾

☒ select all 50 sequences selected[GenPept](#)[Graphics](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	aldehyde dehydrogenase_mitochondrial [Microcebus murinus]	Microcebus murinus	874	874	99%	0.0	99.29%	517	XP_012633825.1
☒	aldehyde dehydrogenase_mitochondrial [Lemur catta]	Lemur catta	872	872	99%	0.0	99.05%	517	XP_045390768.1
☒	aldehyde dehydrogenase_mitochondrial isoform X1 [Eulemur rufifrons]	Eulemur rufifrons	871	871	99%	0.0	98.82%	517	XP_069353277.1
☒	PREDICTED: aldehyde dehydrogenase_mitochondrial [Propithecus coquereli]	Propithecus coquereli	868	868	99%	0.0	98.58%	517	XP_012518934.1
☒	aldehyde dehydrogenase_mitochondrial [Nycticebus coucang]	Nycticebus coucang	865	865	99%	0.0	98.11%	517	XP_053446232.1
☒	aldehyde dehydrogenase_mitochondrial [Otolemur garnettii]	Otolemur garnettii	863	863	99%	0.0	97.64%	517	XP_003796031.1
☒	aldehyde dehydrogenase_mitochondrial [Trachypithecus francoisi]	Trachypithecus francoisi	862	862	99%	0.0	97.64%	517	XP_033076455.1
☒	aldehyde dehydrogenase_mitochondrial [Rhinopithecus roxellana]	Rhinopithecus roxellana	861	861	99%	0.0	97.64%	517	XP_010377208.1
☒	Chain A. Aldehyde dehydrogenase [Homo sapiens]	Homo sapiens	861	861	99%	0.0	97.64%	500	1NZX_A
☒	Homo sapiens aldehyde dehydrogenase 2 family (mitochondrial)_partial [synthetic construct]	synthetic construct	861	861	99%	0.0	97.64%	518	AAP36614.1
☒	aldehyde dehydrogenase_mitochondrial [Nomascus leucogenys]	Nomascus leucogenys	861	861	99%	0.0	97.64%	517	XP_030677272.1
☒	aldehyde dehydrogenase_mitochondrial isoform X1 [Symphalangus syndactylus]	Symphalangus syndactylus	861	861	99%	0.0	97.40%	517	XP_055094692.1
☒	mitochondrial aldehyde dehydrogenase 2 precursor variant_partial [Homo sapiens]	Homo sapiens	861	861	99%	0.0	97.64%	517	BAD97093.1
☒	aldehyde dehydrogenase_mitochondrial [Pongo pygmaeus]	Pongo pygmaeus	860	860	99%	0.0	97.64%	517	XP_054298134.1
☒	Aldehyde dehydrogenase_mitochondrial [Plecturocebus cupreus]	Plecturocebus cupreus	860	860	99%	0.0	97.87%	567	KAL0610756.1
☒	Chain A_MITOCHONDRIAL ALDEHYDE DEHYDROGENASE [Homo sapiens]	Homo sapiens	860	860	99%	0.0	97.64%	494	1CW3_A
☒	PREDICTED: aldehyde dehydrogenase_mitochondrial isoform X2 [Colobus angolensis palliatus]	Colobus angolensis palliatus	860	860	99%	0.0	97.40%	517	XP_011792033.1
☒	aldehyde dehydrogenase_mitochondrial isoform 1_precursor [Homo sapiens]	Homo sapiens	860	860	99%	0.0	97.64%	517	NP_000681.2
☒	aldehyde dehydrogenase_mitochondrial isoform X1 [Pan troglodytes]	Pan troglodytes	860	860	99%	0.0	97.64%	517	XP_016779748.1
☒	aldehyde dehydrogenase_mitochondrial isoform X1 [Theropithecus gelada]	Theropithecus gelada	860	860	99%	0.0	97.40%	553	XP_025256966.1
☒	PREDICTED: aldehyde dehydrogenase_mitochondrial isoform X1 [Colobus angolensis palliatus]	Colobus angolensis palliatus	860	860	99%	0.0	97.40%	553	XP_011792025.1
☒	aldehyde dehydrogenase_mitochondrial [Pan paniscus]	Pan paniscus	860	860	99%	0.0	97.64%	517	XP_003832508.1
☒	aldehyde dehydrogenase_mitochondrial isoform X1 [Macaca fascicularis]	Macaca fascicularis	860	860	99%	0.0	97.40%	557	XP_005572335.4

BLASTX
RESULTS<https://blast.ncbi.nlm.nih.gov/>

NOVEL GENE / PROTEIN

- Our Novel gene is known as *Microcebus ravelobensis* isolate RMR55 scaffold43866 which belongs to the Golden Brown Mouse Lemur
- The novel gene was searched against the Interpro Database
- It is present within the ALDEHYDE DEHYDROGENASE family, which contains over 400,000 members

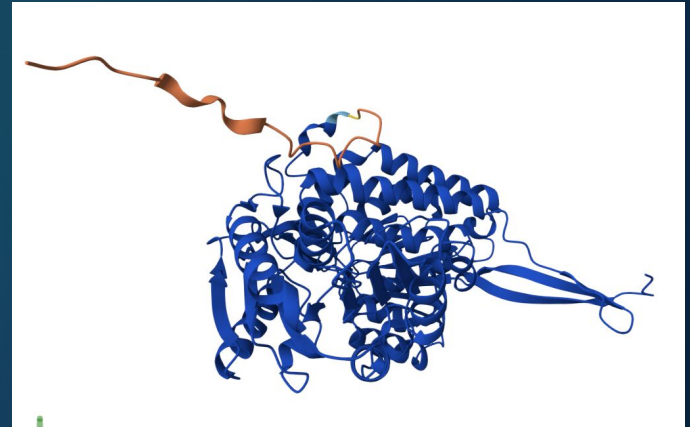
PICTURE OF NOVEL GENE SPECIES



https://media.nature.com/lw1200/magazine-assets/d41586-019-01789-0/d41586-019-01789-0_16776876.jpg

NOVEL GENE / PROTEIN

- The ALDEHYDE DEHYDROGENASE family is typically associated with helping the body convert aldehydes into carboxylic acids, which helps when processing chemicals like alcohol
- Family members of the novel gene are found in almost all organisms



<https://alphafold.ebi.ac.uk/entry/AF-P05091-F1>

MULTIPLE SEQUENCE ALIGNMENT (MSA)

Algorithm	Why?	Parameter Change?
MUSCLE	Accurate and good for proteins	No
CLUSTAL OMEGA	Allows scalable progressive alignment	No

MSA RESULTS

Note: Novel gene is NP_000001.1 and Query gene is NP_000682.1

COLOR SCHEME

clustal2



LEGEND

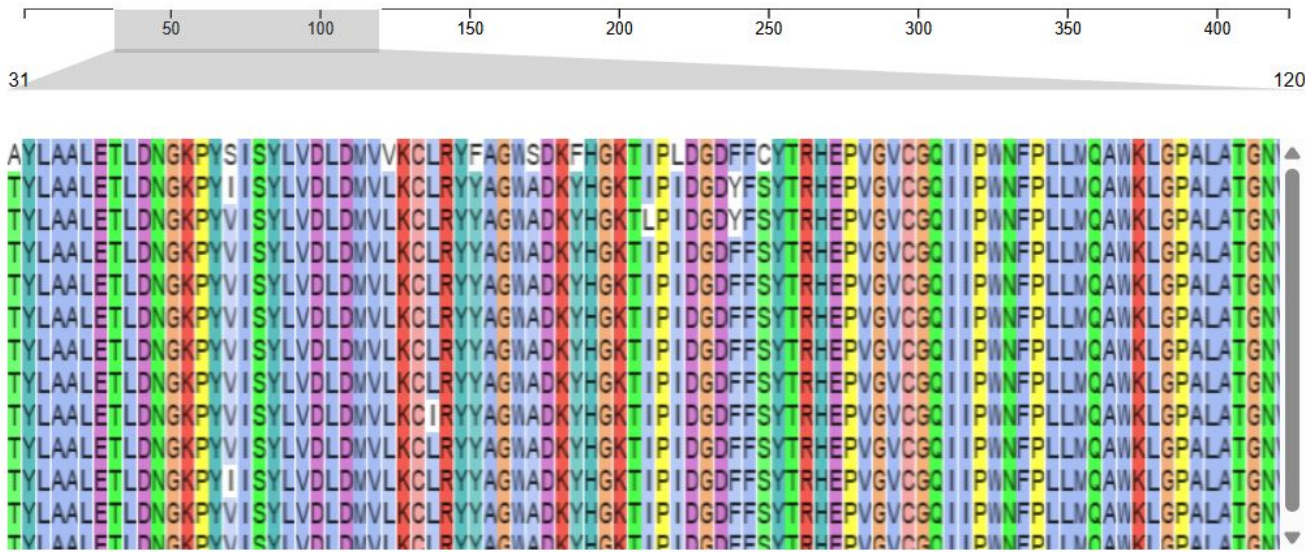
A R N D C Q E G H I L K M F P S T W Y V B X Z



13 sequences



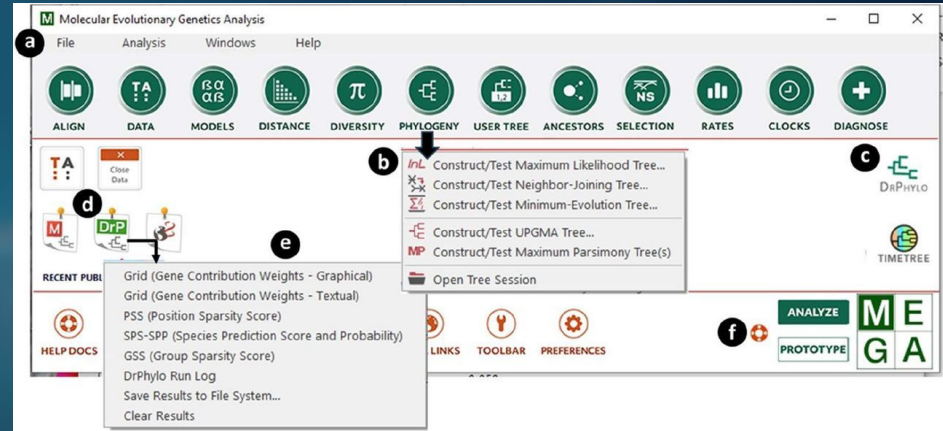
NP_001376401.1
NP_001068835.1
NP_001038076.1
NP_000681.2
NP_001124747.1
VP_000001.1
XP_012633825.1
XP_045390768.1
XP_004753300.2
NP_001421616.1
XP_020033514.2
NP_033786.1
NP_115792.2



MSA CONTINUED...

- Overall both results showed our novel gene shared a great deal in common with other genes from different organisms
- From testing, we found that MUSCLE and Clustal Omega gave similar results
- MUSCLE was selected due to iterative refinement process which provides a higher accuracy compared to Clustal Omega

MOLECULAR EVOLUTIONARY GENETICS ANALYSIS (MEGA) version 12

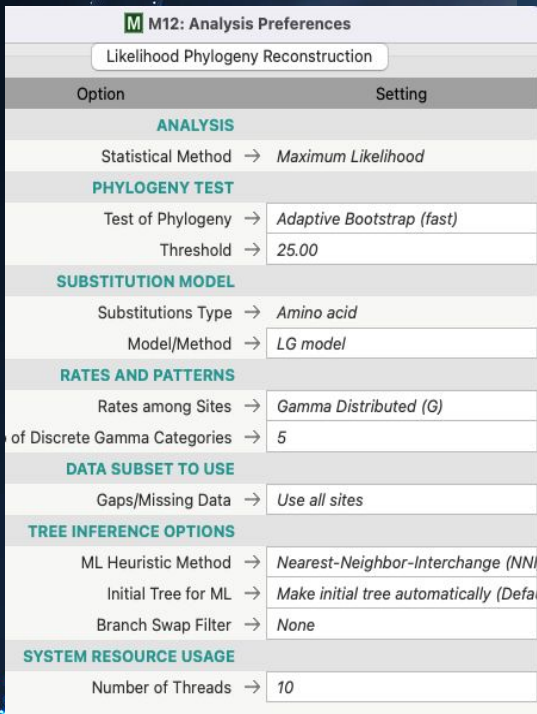


<https://www.biorxiv.org/content/10.1101/2024.12.10.627672v1.full>

PHYLOGENETICS

MEGA's "Find Best DNA/Protein Models (ML)":

- *LG Model + Gamma Distribution (G)*
- *General Reversible Chloroplast (cpREV) + G*
- *LG model + Has Invariant Sites (I)*
- *All Default Parameters*
- *No. of Discrete Gamma Categories = 5*
- *Adaptive Bootstrap (fast) = 25 replicates*



M12: Analysis Preferences

Likelihood Phylogeny Reconstruction

Option	Setting
ANALYSIS	
Statistical Method →	Maximum Likelihood
PHYLOGENY TEST	
Test of Phylogeny →	Adaptive Bootstrap (fast)
Threshold →	25.00
SUBSTITUTION MODEL	
Substitutions Type →	Amino acid
Model/Method →	LG model
RATES AND PATTERNS	
Rates among Sites →	Gamma Distributed (G)
of Discrete Gamma Categories →	5
DATA SUBSET TO USE	
Gaps/Missing Data →	Use all sites
TREE INFERENCE OPTIONS	
ML Heuristic Method →	Nearest-Neighbor-Interchange (NNI)
Initial Tree for ML →	Make initial tree automatically (Default)
Branch Swap Filter →	None
SYSTEM RESOURCE USAGE	
Number of Threads →	10

PHYLOGENETIC TREES

cpREV + I



LG + I



PHYLOGENETIC

Tree

Best model: *LG Model + Gamma Distributed*

Bootstrap @ Novel Clade = 32
- Little statistical significance for clade ($p > 0.05$)

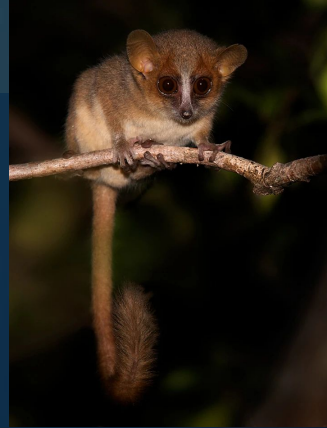
Table. Maximum Likelihood analysis of substitution models

Model	Parameters	BIC	AICc	InL
LG+G	24	4164.430	4005.896	-1978.839
cpREV+G	24	4167.455	4008.922	-1980.352
LG+I	24	4168.309	4009.775	-1980.778



CONCLUSION

- The conserved ALDH2 gene across lemurs demonstrates that this essential detoxification enzyme has remained functionally stable throughout species evolution to protect cells from metabolic toxins.
- Despite the commonality of the family, there are still various examples of un-annotated instances of this important protein
- Bioinformatics offer a variety of powerful tools that can help you investigate different genotypes and phenotypes.



QUESTIONS?